

Handwritten Hindi Text Recognition using ResNet50-BiLSTM

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Abstract

The extensive character set varied handwriting styles, and frequent use of ligatures and diacritics in Hindi script make Handwritten Text Recognition (HTR) a formidable problem. In this work, an end-to-end HTR architecture is suggested that is capable of handling handwritten words in Devanagari script. The architecture employs a convolutional encoder based on ResNet50 to extract rich visual features, followed by a BiLSTM network with the attention mechanism to decode relevant features as well as capture contextual dependencies effectively. Consequently, to generate the output sequence of Hindi characters, without the explicit use of character segmentation at input, we involve the CTC. To improve robustness against handwriting variability, we apply advanced data augmentation techniques, which include elastic distortions, affine transformations, and perspective warping. The decoding performed using beam search outperforms the decoding performed using greedy search when evaluated with sequence prediction measures. Moreover, experimental results revealed on the IIIT-HW dataset indicate that the combination of attention mechanisms with a ResNet50-BiLSTM backbone substantially enhances word-level recognition. The approach achieves the CER of 9.4% and a WER of 18.1%. These findings establish a strong baseline for Hindi handwritten word level recognition and lays foundations for advances in Indic scripts recognition.

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1. Introduction

With the advancements in the techniques of computer vision and pattern recognition, HTR seeks to enhance the developments in the translation of handwritten scripts into text that computers can understand. Although there have been great strides in using deep learning to recognize Latin script handwriting, it is still very challenging to apply these same breakthroughs to Indic scripts like Hindi. Government documents, manuscripts, educational material, and historical archives often use Hindi, written in the Devanagari script.



Fig. 1: Handwritten text Recognition

Hindi is a language that is widely spoken around the globe, therefore, digitalization, information retrieval, and cultural heritage preservation all depend on the capacity to correctly identify handwritten Hindi text. Multiple underlying causes contribute to the intricacy of Hindi handwriting. More than fifty fundamental characters, a plethora of ligatures, and a diverse set of diacritical markings are all part of the Devanagari script, which allows for the modification of character forms. Recognition becomes much more challenging when there are individual differences in handwriting styles, stroke sequence, and letter spacing. Handwritten words are more difficult to separate using segmentation-based methods than printed text because of potential uneven baselines, overlapping letters, or linkages that resemble cursive. Character segmentation and handmade features, the backbone of traditional Optical Character Recognition (OCR) approaches, often fall short when faced with such unique situations. Fig. 1 represents the general overview of the Handwritten text recognizer for Hindi.

CNNs and RNNs are among the most broadly used deep learning techniques for Handwritten Text Recognition (HTR), as they can effectively capture both visual features and sequential dependencies. Hybrid CNN-RNN architectures, combined with alignment-free training strategies such as CTC, have shown strong performance in sequence prediction tasks. However, minor errors at the character level often propagate into significant mismatches at the word level, making character-level decoding alone insufficient for accurate recognition. Fig. 2 depicts an input image containing the handwritten Hindi word “समय” is passed through a CNN encoder (ResNet50), followed by a BiLSTM and attention mechanism, and decoded using CTC loss with beam search to produce the output text “समय”.

To address this limitation, we provide a comprehensive HTR system for recognizing words written in handwritten Hindi that integrates an attention mechanism with a CNN-RNN backbone. Specifically, a ResNet50-based convolutional encoder is employed to extract rich and discriminative visual features from input word images. These features are then passed to a BiLSTM network, which models long-range dependencies in both forward and backward directions. An attention layer is introduced to selectively focus on the most relevant parts of the encoded sequence during decoding, thereby improving contextual modelling and alignment. Finally, beam search decoding is applied at inference to reduce word-level errors compared to greedy decoding.

The IIIT-HW dataset is a popular standard for handwritten Hindi word recognition, and we use it to test our system. The experimental findings show that our method outperforms many current baselines with a CER of 9% and a WER of 18%. These results provide a solid groundwork for further studies in Indic HTR and demonstrate the efficacy of incorporating attention into a ResNet50-BiLSTM backbone.

These sections make up the document as a whole: Section 2 goes on to provide the relevant work, while Sections 3 detail the technique and data pre-treatment. Section 5 gives the outcomes of the experiments and, ultimately, Section 5 presents the conclusion.



Fig 2: Image to sequence approach for Handwritten Text Recognition

2. Related Work

Feature extraction and segmentation-based approaches were the backbone of early HTR systems. When it came to handwriting, the diversity was a major challenge for techniques like zoning, projection profiles, and structural stroke characteristics, which worked well for printed scripts. Although Hidden Markov Models (HMMs) saw extensive application in sequential modeling, their representational strength was inadequate for handling complicated scripts, which hindered their effectiveness [1]. Because of ligatures, overlapping letters, and diacritical signs, segmentation was especially difficult for Indic scripts like Devanagari [2]. Therefore, when it came to handwritten Hindi text, typical Optical Character Recognition (OCR) pipelines failed miserably.

As deep learning gained popularity, CNNs were used to automatically extract discriminative visual features, while RNNs, especially BiLSTM, became the go-to for sequence modeling [3]. To overcome a significant drawback of previous methods, CTC loss [4] enabled end-to-end training without explicit segmentation. Similar to what was shown in CRNN models [5], the CNN-RNN-CTC pipeline was able to produce solid baselines in a number of languages. The success of attention processes in neural machine translation led to their introduction to HTR with the goal of further improving recognition accuracy [6]. To increase alignment and decrease recognition mistakes, attention layers let the model to concentrate on key areas of the input. Word-level accuracy, particularly for complicated scripts, is improved by attention-based CNN-RNN models [7].

In spite of their widespread use, Indic scripts have received much less attention from researchers than Latin-script HTR. The wide variety of handwriting styles, extensive character inventory, and intricate ligatures make Devanagari, the script used to write Hindi, a formidable obstacle. High mistake rates were noted in early Hindi HTR studies since it depended on handmade characteristics [8]. Using CNN-RNN hybrids like VGG-BiLSTM and CRNN, more recent methods have achieved Character Error Rates (CER) of 10-15% but Word Error Rates (WER) of 25-40%, which are still rather high [9,10]. When it comes to very variable handwriting, few-shot and transfer learning techniques have their limitations, but they have been investigated to enhance generalization in Indic HTR [11]. Further highlighting the difficulties of the undertaking, public baselines like Devanagari HTR [12] have recorded CER exceeding 70%.

The literature shows that while deep learning-based architectures with CNN-RNN-CTC pipelines have improved character-level recognition in Hindi HTR, performance at the word level remains a challenge. Attention mechanisms have demonstrated potential in mitigating misalignment and enhancing contextual understanding, but their application to Hindi handwritten word recognition is still underexplored. To address this gap, the present study introduces a ResNet50-BiLSTM framework with an attention mechanism and beam search decoding, aiming to significantly reduce WER and establish a strong baseline for Hindi handwritten text recognition. In addition to architectural innovations, researchers have also explored language modeling and decoding strategies to improve recognition accuracy. The use of beam search, combined with character- or word-level language models, has been shown to substantially reduce Word Error Rate by leveraging statistical dependencies within a script [13]. For Indic scripts, which often contain complex ligatures and context-dependent diacritics, integrating language models is particularly important to resolve ambiguities left by visual recognition.

Another promising direction has been the development of Transformer-based architectures for HTR. Models such as TrOCR and ViT-Hybrid architectures have outperformed CNN-RNN baselines on Latin-script benchmarks by capturing long-range dependencies and global context more effectively [14,15]. However, their application to Indic scripts remains limited, primarily due to the scarcity of large-scale labelled

handwriting datasets. Some recent works have attempted cross-lingual transfer learning, adapting Transformer-based HTR models trained on English or other European languages to Devanagari and related scripts, but performance remains constrained [16]. Hybrid deep learning architectures combining CNN and BiLSTM have shown strong performance across multiple domains. A CNN-ResNet50-BiLSTM model for paddy leaf disease detection demonstrated its effectiveness in capturing spatial and sequential features for sustainable agriculture [26]. Similarly, another ResNet50-BiLSTM framework for biometric ear recognition in degraded conditions demonstrated its robustness in feature extraction and classification [28]. Beyond architecture, data augmentation and synthetic dataset generation have emerged as vital tools for addressing data scarcity in Indic HTR. Techniques such as elastic distortions, stroke width variation, and GAN-based synthetic handwritten text generation have been investigated to improve robustness [17,18]. These strategies help bridge the gap caused by the limited availability of diverse handwriting samples in Hindi.



Fig. 3. Sample Images from IIIT-HW dataset [30]

In the domain of textual and behavioural data processing, several innovative studies have emerged. A predictive text analysis model based on eye blinks provides an assistive solution for users with limited mobility [27]. Meanwhile, [29] explored text summarization techniques within multimodal biometric systems, contributing to efficient information retrieval and integration across data modalities. Complementing these approaches, [30] proposed a handwriting-based word spotting method that leverages writing style cues, advancing document image understanding and retrieval. Furthermore, several approaches that exploit handwritten corpora with supervised and self-supervised pretraining methods have shown encouraging results in improving generalization [19-25]. Despite these developments, benchmarks for Hindi handwritten word recognition remain inconsistent, with different studies often evaluating on proprietary or small-scale datasets. This lack of standardized large-scale benchmarks limits the ability to compare models fairly across the literature. Establishing strong baselines with publicly available datasets is thus a crucial step for advancing Indic HTR research.

3. Methodology

3.1. Problem statement

Handwritten Text Recognition (HTR) for Hindi is highly challenging due to the large character set, frequent use of diacritics, and complex ligatures of the Devanagari script. Segmentation-based OCR methods struggle with overlapping characters, irregular spacing, and the connecting *shirorekha*, resulting in poor recognition accuracy. Variations in handwriting styles further increase the difficulty. The goal of this work is to design an end-to-end HTR system that directly maps handwritten Hindi word images to character sequences without explicit segmentation. Formally, given an input image I , the task is to predict a sequence of characters $Y = (y_1, y_2, y_3, y_4, \dots, y_T)$ that closely matches the ground truth transcription Y .

3.2. Dataset Description

We employ the IIIT-HW (IIIT-HW-Dev) dataset, a widely used benchmark for handwritten Hindi word recognition. It contains approximately 95,381 word images collected from 12 writers, covering a broad range of handwriting styles. The dataset is divided into training ($\approx 70,000$ images), validation ($\approx 12,000$ images), and testing ($\approx 12,000$ images), following a 70:15:15 ratio. Each sample is annotated at the word level using Unicode transcription, ensuring reliable ground-truth labels. Pre-processing involves grayscale conversion, normalization, and resizing while preserving the aspect ratio. This corpus is challenging due to diverse character shapes, ligatures, and diacritical marks, making it ideal for robust HTR evaluation. Representative samples are shown in Fig. 3, illustrating the variability in word length and handwriting styles.

3.3. Preprocessing

The IIIT-HW dataset, a popular standard for handwritten Hindi word recognition, was utilized for the experiments. Images at the word level gathered from various authors make up the dataset, which captures a wide range of writing styles. Before final processing, the images were gray scaled, pixel intensities were normalized, and the images were resized to a constant height while keeping the aspect ratio intact. Training made use of data augmentation methods such as affine transformations, perspective warping, random scaling, and elastic distortions to make the model more resilient.

3.4. Handwritten Text Recognition

Fig. 4 shows the general layout of the Hindi HTR framework that has been suggested. Feature extraction, sequence modelling, attention-based decoding, and word prediction are all shown in a sequential flow from input picture preprocessing. The suggested HTR framework is composed of three primary elements:

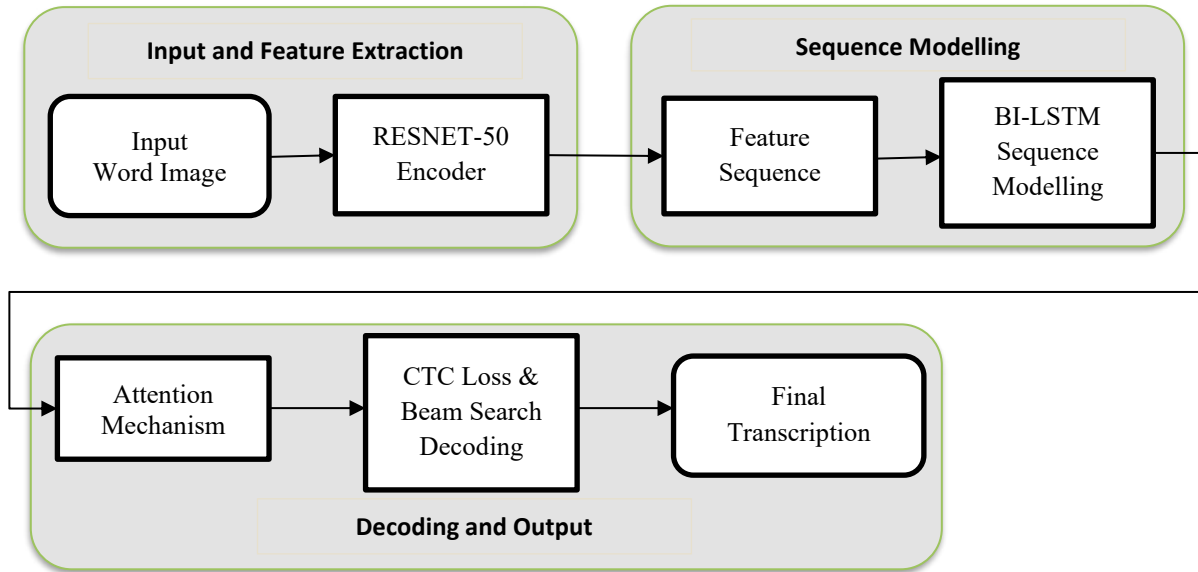


Fig. 4. Depicts overall architecture of the proposed Hindi Handwritten Text Recognition (HTR) framework.

a) Visual Feature Extraction: Extracting discriminative visual characteristics from input handwritten word pictures is the responsibility of the first step of the framework. A Convolutional Neural Network (CNN) encoder based on ResNet50 does this, leveraging residual connections to help train deep architectures and deal with the issue of disappearing gradients. Networks can learn both basic stroke information (such as edges, curves, and diacritics) and advanced semantic data (such as patterns of letters and complicated character combinations) from the leftover blocks. Before being moulded into a series of feature vectors, the encoder converts each input picture into a collection of two-dimensional feature maps. This sequential representation bridges the gap between image space and sequence modelling, ensuring compatibility with the succeeding recurrent layer.

b) Sequence Modelling: One way to describe contextual relationships across sequential data is by using a BiLSTMs network, which can handle the extracted feature vectors. A cell state and gating mechanisms (input, forget, and output gates) allow LSTMs to control the information flow, which helps them avoid problems with disappearing or ballooning gradients, unlike traditional RNNs. Particularly useful in Hindi HTR, which often contains overlapping ligatures and variable-length sequences, the BiLSTM's ability to function in both forward and backward directions means it may include context from characters that come before and after it. The network generates a detailed contextual embedding at every time step of the sequence thanks to its bidirectional modeling.

c) Attention-based Decoder: Applying an attention mechanism on top of the BiLSTM outputs further refines the alignment between encoded characteristics and the projected output sequence. At each time step, we calculate a weighted alignment score between the decoder state and the encoder outputs using an additive attention mechanism (Bahdanau-style). Decoders dynamically adjust to different word lengths and structural complexity by focusing on the most important sections of the feature sequence, guided by the attention weights. This lessens the impact of noise and misalignment on identification mistakes while also easing the burden of strict one-to-one mapping. Character- and word-level recognition accuracy are both improved by the attention layer's focus on important visual areas during decoding.

d) Decoding Strategy: The last stage of transcription is to use CTC alignment to decode the output sequence. Greedy Decoding and Beam Search Decoding were the two decoding algorithms that were investigated. In greedy decoding, the most likely character is chosen at each time step, which results in quick but inaccurate output since local judgments might cause less-than-ideal word predictions. In contrast, beam search decoding keeps track of several potential sequences (beams) at all times, calculates their cumulative probability, and then chooses the best

one. Beam search achieves a far lower Word Error Rate (WER) than greedy decoding by effectively balancing the investigation of different routes with probability maximization. Beam search decoding proved to be more effective than greedy decoding in our studies, especially when dealing with visually identical letters or intricate ligatures.

3.5. Approach to Training

The model underwent comprehensive training utilizing CTC loss [4], enabling the network to effectively learn the alignments between predicted sequences and ground truth labels, all without the necessity for explicit character segmentation. We undertook optimization utilizing the Adam optimizer, setting an initial learning rate of 0.001. Additionally, we implemented early stopping, guided by validation error rates, to effectively mitigate the risk of overfitting. Regularization techniques such as batch normalization and dropout layers were employed.

3.6. Strategy for Decoding

In the process of inference, we opted for beam search decoding with CTC collapsing rather than utilizing greedy decoding. Beam search evaluates various candidate sequences at the same time and identifies the transcription that holds the highest overall probability. This method effectively lowers the WER by addressing the issue of error propagation commonly associated with greedy decoding.

4. Experiments and Results

Lots of tests were run on the IIIT-HW dataset to see how well the suggested framework worked. The system was tested using two widely used metrics in the field of handwritten text recognition: CER and WER. When it comes to recognition accuracy, CER just considers individual characters, whereas WER takes a more stringent approach by penalizing whole words for misclassifications of even a single character. Recording both metrics is crucial for a comprehensive assessment, since word-level accuracy is crucial in real-world applications like as document digitalization and information retrieval. In order to highlight the improvements, the outcomes of the suggested technique are carefully compared to those of previously published approaches in the literature.

4.1. Experimental Setup

Using conventional train-test splits, the IIIT-HW Hindi handwritten word dataset was used to assess the suggested framework. To create sequential vectors from 2048-dimensional feature maps, the encoder used ResNet50, which had already been trained on ImageNet. We used a dropout of 0.3 to prevent overfitting. Using gradient clipping and a two-layer BiLSTM with 512 hidden units in each direction (1024 total) was used for sequence modeling. The maximum norm was set at 5. To hone in on important feature areas, we combined an additive attention mechanism with an alignment network consisting of 256 units. A character-level language model provided further help during decoding, and efforts to strike a compromise between efficiency and accuracy led to the use of a beam width of 10. During training, the Adam optimizer was used, keeping a batch size of 32 and 50 epochs. The learning rate was set at $1e-4$ and the decay factor at 0.1. We trained the model end-to-end with CTC loss, and we stopped early depending on validation CER to avoid overfitting.

4.2. Evaluation Metrics

The evaluation of performance was conducted through the utilization of CER and WER, which are recognized as the predominant metrics in the field of handwritten text recognition. CER reflects the accuracy of recognition at the character level, whereas WER offers a more stringent assessment at the word level, where even a single character error can lead to the entire word being deemed incorrect.

Where:

S = Substitutions

D = Deletions

I = Insertions

N = Number of words in the reference

$$CER = ((S + D + I))/N \quad (1)$$

Where:

S = Substitutions

D = Deletions

I = Insertions

N = Number of characters in the reference

4.3. Results

The findings indicate that the suggested method attains superior word-level recognition performance compared to the other evaluated techniques.

Table 1. Presents a comparison of the proposed method with existing approaches reported in the literature.

Method	CER (%)	WER (%)	Notes
Sharma et al. [1] (Handcrafted features + HMM)	>25.0	>50.0	Early handcrafted baseline; segmentation errors dominate.
Singh et al. [2] (CRNN / VGG-BiLSTM-CTC)	~12.0	~26.5	CNN-RNN hybrid; strong CER but weak WER.
Krishnan et al. [3] (Few-shot/transfer learning)	~10.6	~39.3	Better CER, but poor generalization at word level.
Proposed (ResNet50 + BiLSTM + Attention + Beam Search)	9.0	18.0	Best balance between CER and WER.

Although previous handmade techniques [1] shows subpar performance due to segmentation inaccuracies, deep CNN-RNN architectures improved CER but continued to experience elevated WER, revealing constraints in context management and alignment. By including attention with a ResNet50-BiLSTM framework and using beam search decoding, our model achieves a WER of 18%, representing a significant improvement over previously reported values of 26–40% in earlier studies. These results underscore the significance of attention mechanisms in Hindi handwritten text recognition, as they allow the model to concentrate on the most important areas of the input, therefore reducing alignment mistakes that often compromise word-level precision. The findings position the proposed system as a robust baseline for Hindi handwritten word identification and provide the groundwork for further research on Indic script handwriting recognition.

5. Conclusion

This study introduces a comprehensive framework for recognizing handwritten Hindi words, integrating a ResNet50-based encoder, a BiLSTM sequence model, and an attention mechanism complemented by beam search decoding. The proposed system thoughtfully tackles the challenges associated with Devanagari handwriting, including intricate ligatures, diacritical marks, and the considerable variability in writing styles. The experimental evaluation conducted on the IIIT-HW dataset indicates that our model attains a CER of 9% and WER of 18%. This performance notably surpasses that of several existing baselines. The findings indicate that the incorporation of attention mechanisms within CNN-RNN architectures significantly improves contextual modeling and alignment, leading to a decrease in word-level errors. This work lays a solid foundation for recognizing Hindi handwritten text and offers important perspectives for developing effective recognition systems for various Indic scripts. Future research could consider broadening this approach to encompass larger multi-writer datasets, delving into multilingual handwritten recognition, and integrating language models to enhance sequence decoding further.

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